
Figure 1: Features of Gadfly

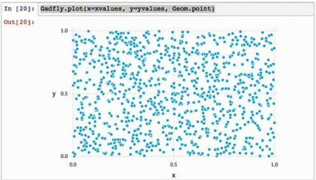


Figure 2: Gadfly - a simple plot

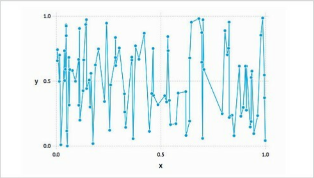


Figure 3: Gadfly - generating a line graph

```
xvalues = rand(100)
yvalues = rand(100)
Gadfly.plot(x=xvalues, y=yvalues, Geom.point, Geom.line)
```

The smoothened version is generated with *Geom.smooth*, as shown in Figure 4.

The plot can be customised with details such as *axis title* with the following code:

```
Gadfly.plot(x=1:10, y=2.^rand(10),
Scale.y_sqrt, Geom.point, Geom.smooth,
Guide.xlabel("Stimulus"), Guide.ylabel("Response"), Guide.
title("Dog Training"))
```

The output of the code is shown in Figure 5.

## Gadfly: Histogram with RDataSets

Gadfly enables the creation of histograms effortlessly. A sample histogram with RDataSets is shown in Figure 6.

```
using RDatasets
Gadfly.plot(dataset("car", "SLID"), x="Wages",
color="Language", Geom.histogram)
```

## Multilayer plots

Gadfly provides options to layer and stack plots. A sample code to layer a point plot and line plot is shown below:

```
Gadfly.plot(layer(x=rand(10), y=rand(10), Geom.point,
order=1),
     layer(x=rand(10), y=rand(10), Geom.line, order=2))
```

Here, the keywords *layer* and *order* are used to provide multilayer plots in a specified order.

## Gadfly backends

Gadfly provides support to write to various backgrounds such as SVG and SVGJS. The support for SVG and SVGJS is provided by default. However, the other backends such

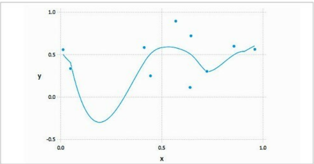


Figure 4: Gadfly – Geom.smooth

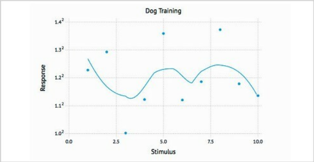


Figure 5: Gadfly plot with axis title information

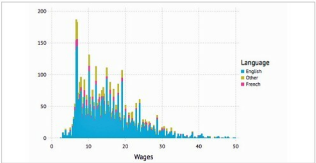


Figure 6: A sample histogram